

AIBEL JOMON

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OBJECTIVE

Seeking experience through internships to develop as an engineer. Looking for design or control system internship roles.

EDUCATION

Texas A&M University (TAMU), College Station, Texas Expected Graduation: May 2029
General Engineering – pursuing Bachelor of Science in Mechanical Engineering and Electrical Engineering Minor
Relevant Coursework: Engineering Computation, Newtonian Mechanics, Engineering Lab I, Calculus I & II
Major GPA: 4.0, Cumulative GPA: 4.0

Port Neches – Groves High School, Port Neches, Texas August 2021 - May 2025
Weighted GPA: 4.54, Unweighted GPA: 4.0

RELEVANT EXPERIENCE

TAMU SAE Baja, *Controls Sub-Team Member*, College Station, Texas April 2026 - Current

- Completed Baja Design Challenge – Create SAE 2026 Rules legal brake pedal.
- Designed Brake out of **6061-T6 Aluminum** to handle **2,200+** lbf stomp force, exceeding 450 lbf requirement by **388%**.
- Validated structural integrity through SolidWorks linear static **FEA**, engineering an I-beam profile that maintained Factor of Safety of **4.97** while only weighing **1.03 lbs**.
- Wrote **White Paper** documenting Design and Validation process and Technical Drawings for Pedal design.

Naval Design Team, *Electric Boat Sub-Team Member*, College Station, Texas August 2025 - Current

- Created 3D-printed custom mounting brackets for high-current speed controllers and transmitters used for **Promoting Electrical Propulsion (PEP)** Competition boat using Fusion360, prioritizing **Design for Assembly (DFA)** principles.
- Configured transmitter for **differential thrust**, optimizing boat steering radius and implemented a kill switch.
- Configured receiver to send signals to throttle motors, achieving **~40 lbs** thrust in water at 50% power output in tests.

Athletes (Event & Training Center), *Event Assistant*, Port Neches, Texas May 2023 - August 2024

- Coordinated setup of weekend events, ensuring customer satisfaction and strict schedule adherence.
- Prioritized urgent customer requests and facility maintenance, acting as primary point of contact for clients to ensure a seamless experience.

PROJECTS & ORGANIZATIONS

Temperature Sensor and Microprocessor Basics, *Personal Engineering Project* January 2025 - Current

- Utilized **Steinhart-Hart Equation** to find unique coefficients for **NTC thermistor**.
- Developed C firmware in the **ESP-IDF** framework to monitor ambient temperature and achieve $\pm 0.2^{\circ}\text{C}$ accuracy.
- Optimized signal integrity using hardware decoupling with **0.1 μF capacitors**.
- Worked with **C** and **Rust** on Microcontrollers (ESP32, STM32, Raspberry Pi Pico) to implement embedded skills.

American Society of Mechanical Engineers (ASME), *Member* August 2025 - Current

- Attended monthly meetings, engaged with industry representatives to analyze current engineering challenges.

Remote Operated Vehicle, *Personal Engineering Project* July 2025 - September 2025

- Integrated **ESP32** with DC motor drivers and 5V motors by designing a custom power distribution circuit.
- Coded embedded firmware in **MicroPython**, integrating **Pulse Width Modulation (PWM)** for motor control.

TECHNICAL SKILLS

- **CAD & Design:** SolidWorks, Fusion 360, FEA, Design for Manufacturing (DFM).
- **Electronics & Hardware:** Microcontrollers (ESP32, STM32, Raspberry Pi, Arduino), Soldering, Circuit Multimeter Diagnostics, I2C protocols, ESP-IDF framework.
- **Programming:** Python, Embedded C, Rust, Linux.
- **Manufacturing:** 3D printing (FDM), Rapid Prototyping, Stick Welding Fundamentals.